

Qatar-1b

R-2743

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_200801 · created 2026-09-02T01:38:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459061.3741 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.056
Transit depth (Rp/Rs)^2	3.69 +/- 2.13 [%]
Orbital Inclination (inc)	86.87 +/- 2.82
Airmass coefficient 1 (a1)	0.023 +/- 0.011
Airmass coefficient 2 (a2)	2.92 +/- 1.51
Scatter in the residuals of the lightcurve fit is	38.79 %
Best Comparison Star	None
Optimal Aperture	1.37
Optimal Annulus	5.32
Transit Duration (day)	0.0308 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.056 vs 0.1463 (deviation 0.7σ)
Duration fit vs published	0.0308 d vs 0.0692 d (deviation 6.1σ)
Mid-time O–C	-2041.5 min
Depth SNR	2.9
Residual scatter	38.79 %

Light curve

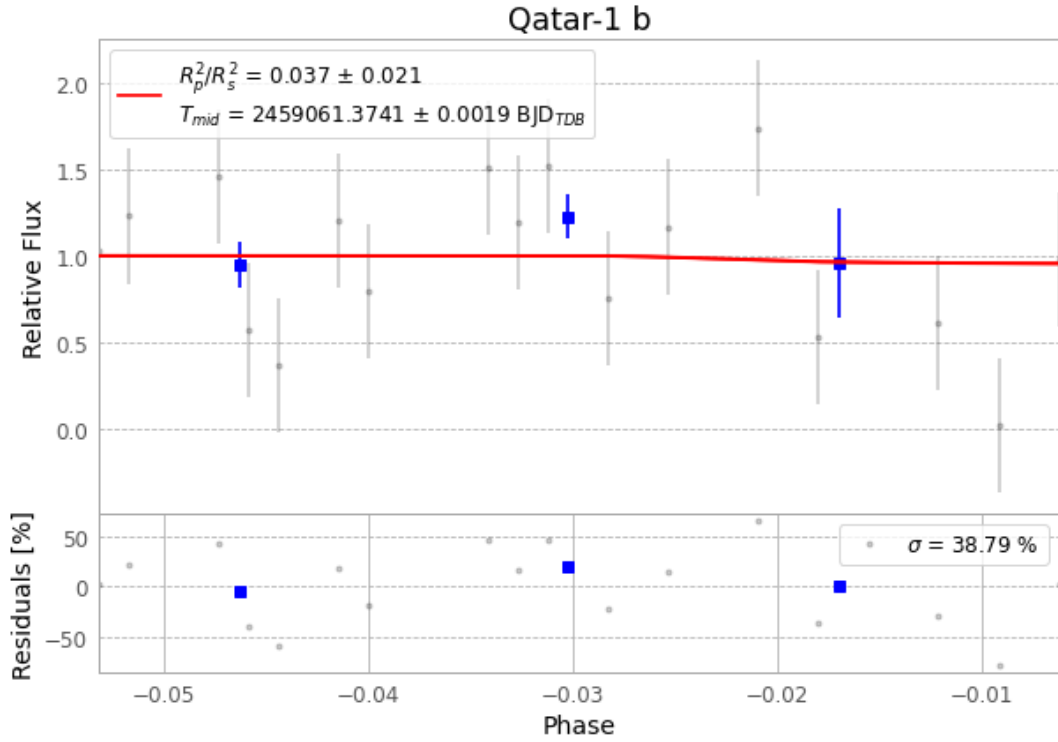


Figure 1 — FinalLightCurve_Qatar-1 b_2020-07-31.png (SHA-256 2068206b05559462...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [594, 157], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c0940b55fa7e07ab...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

36 FITS frames (24 MB), Qatar-1200801051212.FITS → Qatar-1200801065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-200801.log (01fda68f3d47...), FinalLightCurve_Qatar-1 b_2020-07-31.png (2068206b0555...), FinalLightCurve_Qatar-1 b_2020-07-31.csv (54e39705051a...)

Compute resources

Wall time 252.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:38Z.